

# An Improved Computing Model for a Two-wheeled Self-balancing Vehicle's State Determination

Sufeng Wang

School of Computer Science, National University of Defense Technology  
Changsha, China

Hongyi Lu, Fangyong Hou

State Key Laboratory of High Performance Computing, National University of Defense Technology  
Changsha, China

**Abstract**—Owing to the estimating accuracy of state variable for a two-wheeled self-balancing vehicle is not high in a transient status, this paper presents an improved computing model based on a simplified force model of a two-wheeled self-balancing vehicle. Experimental results show that the improved computing model reduces the estimating error of state variable for a two-wheeled self-balancing vehicle in a transient status.

**Keywords**—two-wheeled self-balancing vehicle; improved computing model

## I. INTRODUCTION

In the field of mobile robotics research, a two-wheeled self-balancing vehicle is one of the important research directions. A two-wheeled self-balancing vehicle is a nonlinear, nature unstable and prone to interference of dynamical system.

The idea of balancing a vehicle on two wheels is based on the concept of an inverted pendulum model. Felix Grasser etc., from the Swiss Federal Institute of Technology, developed the two-wheeled self-balancing robot JOE based on the control principle of inverted pendulum that be able to better achieve dynamic equilibrium [1]. Tu Yunwu etc., from University of Science and Technology of China, has developed a two-wheeled self-balancing vehicle: *Free Mover* [2][3]. *Free Mover* is a two-wheeled self-balancing electric vehicle using a parallel structure. A two-wheeled self-balancing vehicle has evoked a lot of interest recently and at least one commercial product.

In these two-wheeled self-balancing vehicles [1][2], the estimating model of state variable regards generally the change of force as a noise, and state variable can be corrected by the acceleration sensor. In a steady status or a small change of the speed, the estimating accuracy of state variable is relatively high, and a two-wheeled self-balancing vehicle runs smoothly. However, when in the accelerating status, decelerating status or substantial change of the speed, the estimating accuracy of state variable is poor, and the two-wheeled self-balancing vehicle does not run smoothly.

In this paper, based on the force analysis of a two-wheeled self-balancing vehicle, we derive a simplified model of the force, which regards no longer completely the change of force as a noise. Although the amount of computing increases, the estimating accuracy of state variable for a two-wheeled self-balancing vehicle is relatively high in a transient status.

## II. DYNAMICS MODEL OF A TWO-WHEELED SELF-BALANCING VEHICLE

Consider the linear motion. As shown in Fig. 1.

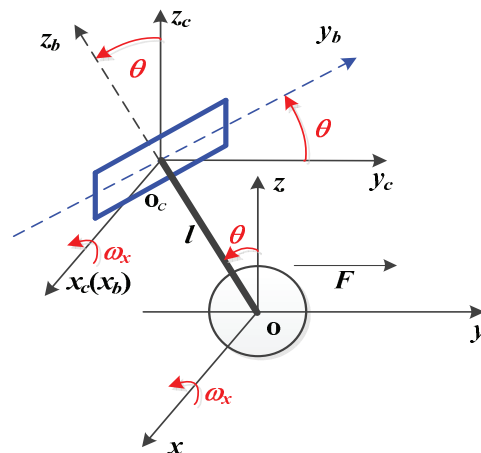


Fig. 1 On mechanical analysis of a two-wheeled self-balancing vehicle in a linear motion

Assume the distance from the center of gravity ( $O_c$ ) of the vehicle body (rotator) to the center of rotation ( $O$ ) is  $l$ .

The origin of the body coordinate system is in the center of gravity ( $O_c$ ) of the vehicle body.  $x_b$  refers to the right of the vehicle body along the horizontal axis, and  $y_b$  points along the front of the vehicle body, and  $z_b$  refers to the sky along the vertical axis of the vehicle body to meet the right-hand rule.

The origin of gravity inertial coordinate system

clicks in the center of gravity ( $O_c$ ) at the origin of the body coordinate system.  $y_c$  axis points forward movement, and  $z_c$  axis points along the vertical direction toward the sky, and  $x_c$  axis is determined in accordance with the right-hand rule.

The coordinate system of a two-wheeled self-balancing vehicle taps in the center of rotation ( $O$ ) at the center of the wheel. So  $y$ -axis points forward movement (with the  $y_c$  axis in the same direction),  $z$ -axis along the vertical direction toward the sky (with the  $z_c$  axis in the same direction), and  $x$ -axis being determined according to the right-hand (with the  $x_c$  axis in the same direction).

$m_c$  is the mass of the vehicle body (rotating body), and  $m_w$  is the mass of these wheels.  $F$  is the resultant force generated by the vehicle and resistance force.

A two-wheeled self-balancing vehicle body rotates around the  $x$ -axis, and the vehicle body angular velocity is  $\omega_x$ , and the vehicle body attitude angle is  $\theta$ . The vehicle moves along the  $y$ -axis linearly, and the acceleration is  $a_y$ , the velocity is  $v_y$ , the displacement is  $p_y$ . The acceleration of center of gravity ( $O_c$ ) of the vehicle body is ( $a_{cy}$   $a_{cz}$ ), and also the velocity is ( $v_{cy}$   $v_{cz}$ ).

Since the resultant force  $F$  is generated inside the motor of the vehicle and the resistance force, difficult to get an accurate model, it cannot be seen as an input. Therefore, it is regarded as a state variable in the following equations [4].

$$\begin{cases} F = (m_w + m_c)a_y - m_c l \frac{d^2(\sin \theta)}{dt^2} \\ = (m_w + m_c)a_y - m_c l \cos \theta \frac{d\omega_x}{dt} + m_c l \omega_x^2 \sin \theta \\ \frac{d\omega_x}{dt} = \frac{F \cdot l \cos \theta + m_c g l \sin \theta}{J_c} \end{cases} \quad (1)$$

where  $J_c$  is the moment of inertia of the body (rotating body).

Due to the low velocity of a two-wheeled self-balancing vehicle, the wind resistance is very small, negligible. The kinetic equations are shown as follows:

$$\begin{cases} \frac{d\omega_x}{dt} = c_1 + \Delta T_g \\ \frac{d\theta}{dt} = \omega_x \\ \frac{da_y}{dt} = \Delta F \\ \frac{dv_y}{dt} = a_y \\ \frac{dp_y}{dt} = v_y \end{cases} \quad (2)$$

where  $c_1$  is shown as follows:

$$c_1 = \frac{(m_c + m_w)l a_y \cos \theta + m_c l^2 \omega_x^2 \sin \theta \cos \theta + m_c l g \sin \theta}{m_c l \cos \theta + J_c} \quad (3)$$

This interference  $\Delta T_g$ , which is accidental rotational torque, can be considered as a white noise.  $\Delta F$ , which is the occasional change of a force, can be considered as a noise [1][2].

### III. AN IMPROVED COMPUTING MODEL FOR A TWO-WHEELED SELF-BALANCING VEHICLE'S STATE DETERMINATION

#### A. An Improved Computing Model

The basic control flow of a two-wheeled self-balancing vehicle is shown in Fig. 2. According to Fig. 2 as well as relative knowledge, we can create a complete mathematical model [1], which is too complicated, but we just need a simple model because of the amount of computing.

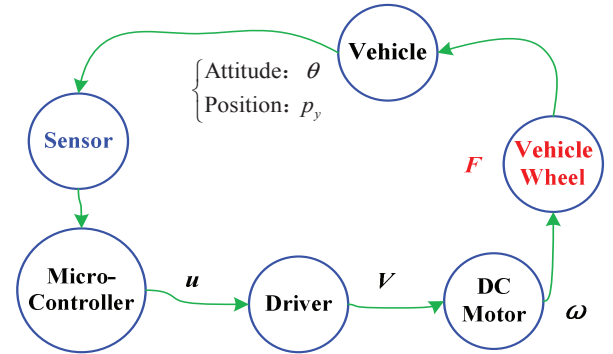


Fig. 2 The basic control process of a two-wheeled self-balancing vehicle

Generally the voltage  $V$  is proportional to the control amount  $u$  (driver linear amplification), and the angular velocity of the DC motor is proportional to the voltage  $V$  (motor equation), and a two-wheeled self-balancing vehicle employs a PID control rule.

The goal of a two-wheeled self-balancing vehicle is to control the attitude angle  $\theta$  tend to 0. If the attitude angle  $\theta$  has a deviation, then a driving force  $F$  is to be generated in the opposite direction to eliminate the bias, so the simple relationship of the driving force with the attitude angle is  $F \propto -\theta$  (i.e.  $F$  is proportional to  $\theta$  but in the opposite direction). Furthermore, the driving force  $F$  is proportional to the acceleration  $a_y$  in  $y$ -axis, that is  $a_y \propto F \propto -\theta$ . So that the Eq.(2) can be corrected to

$$\begin{cases} \frac{d\omega_x}{dt} = c_1 + \Delta T_g \\ \frac{d\theta}{dt} = \omega_x \\ \frac{da_y}{dt} = -k_p \frac{d\theta}{dt} + \Delta F = -k_p \omega_x + \Delta F \\ \frac{dv_y}{dt} = a_y \\ \frac{dp_y}{dt} = v_y \end{cases} \quad (4)$$

where  $k_p$  is a constant coefficient, and  $\Delta F$  is still considered to be a noise, but the model is clearly more accurate and uncomplicated.

### B. The Equation of State

Set the sampling period as  $T$ , and the discrete equations (backward difference) of the Eq. (4) are shown as follows:

$$\begin{cases} \omega_x(k) = \omega_x(k-1) + c_1(k-1) \cdot T + \Delta T_g(k) \cdot T \\ \theta(k) = \theta(k-1) + \omega_x(k-1) \cdot T + 0.5[c_1(k-1) + \Delta T_g(k)] \cdot T^2 \\ a_y(k) = a_y(k-1) - k_p \omega_x(k-1) \cdot T + \Delta F(k) \cdot T \\ v_y(k) = v_y(k-1) + a_y(k-1) \cdot T + 0.5\Delta F(k) \cdot T^2 \\ p_y(k) = p_y(k-1) + v_y(k-1) \cdot T + 0.5a_y(k-1) \cdot T^2 \end{cases} \quad (5)$$

Select  $\mathbf{x} = [\omega_x \quad \theta \quad a_y \quad v_y \quad p_y]^T$  as the state variable of a two-wheeled self-balancing vehicle.

The equation of state is

$$\mathbf{x}_k = \begin{bmatrix} \omega_x(k) \\ \theta(k) \\ a_y(k) \\ v_y(k) \\ p_y(k) \end{bmatrix} = \begin{bmatrix} \omega_x(k-1) + c_1(k-1) \cdot T + \Delta T_g(k) \cdot T \\ \theta(k-1) + \omega_x(k-1) \cdot T + 0.5[c_1(k-1) + \Delta T_g(k)] \cdot T^2 \\ a_y(k-1) - k_p \omega_x(k-1) \cdot T + \Delta F(k) \cdot T \\ v_y(k-1) + a_y(k-1) \cdot T + 0.5\Delta F(k) \cdot T^2 \\ p_y(k-1) + v_y(k-1) \cdot T + 0.5a_y(k-1) \cdot T^2 \end{bmatrix} + \begin{bmatrix} \Delta T_g(k) \cdot T \\ 0.5\Delta T_g(k) \cdot T^2 \\ \Delta F(k) \cdot T \\ 0.5\Delta F(k) \cdot T^2 \\ 0 \end{bmatrix} \quad (6)$$

$$= f(\mathbf{x}_{k-1}) + \mathbf{n}_x(k)$$

$\mathbf{n}_x(k) = [\Delta T_g(k)T \quad 0.5\Delta T_g(k)T^2 \quad \Delta F(k)T \quad 0.5\Delta F(k) \cdot T^2 \quad 0]^T$ .  $\Delta T_g(k)$  is the occasional interference rotational torque, and it is random, so it is seen as a white noise.  $\Delta F(k)$ ,

the occasional change of a force, can be considered as a noise.

Taking into account the small amount of  $\theta$  and  $\omega_x$ ,  $c_1$  of the Eq.(3) is approximately as

$$\begin{aligned} c_1 &= \frac{(m_c + m_w)l a_y \cos \theta + m_c l^2 \omega_x^2 \sin \theta \cos \theta + m_c l g \sin \theta}{m_c l \cos \theta + J_c} \\ &\approx \frac{m_c l g}{m_c l + J_c} \theta + \frac{(m_c + m_w)l}{m_c l + J_c} a_y \\ &= c_{11} \theta + c_{12} a_y \end{aligned} \quad (7)$$

The accelerometer and gyroscope are mounted on the vehicle body. The accelerometer-measured value is  $\tilde{a}_b = (\tilde{a}_{bx} \quad \tilde{a}_{by} \quad \tilde{a}_{bz}) = (0 \quad \tilde{a}_{by} \quad \tilde{a}_{bz})$ . Note: The measured values includes in the gravitational acceleration  $g$ . The gyro-measured value is  $\tilde{\omega}_b = (\tilde{\omega}_{bx} \quad \tilde{\omega}_{by} \quad \tilde{\omega}_{bz}) = (\tilde{\omega}_x \quad 0 \quad 0)$ .

By the characteristics of these sensors, the observation variable is  $\mathbf{z} = [\tilde{\omega}_x \quad \tilde{a}_{by} \quad \tilde{a}_{bz}]^T$ .

$$\begin{bmatrix} a_{by} \\ a_{bz} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} a_{cy} \\ a_{cz} - g \end{bmatrix} \quad (8)$$

Then

$$\begin{cases} a_{by} = a_{cy} \cos \theta + (a_{cz} - g) \sin \theta \\ a_{bz} = -a_{cy} \sin \theta + (a_{cz} - g) \cos \theta \end{cases} \quad (9)$$

As well as

$$\begin{cases} a_{cy} = a_y - l \cos \theta \frac{d\omega_x}{dt} + l \omega_x^2 \sin \theta \\ a_{cz} = -l \sin \theta \frac{d\omega_x}{dt} - l \omega_x^2 \cos \theta \end{cases} \quad (10)$$

Then

$$\begin{cases} a_{cy} = a_y - l c_1 \cos \theta + l \omega_x^2 \sin \theta \\ a_{cz} = -l c_1 \sin \theta - l \omega_x^2 \cos \theta \end{cases} \quad (11)$$

So

$$\begin{cases} a_{by} = (a_y - l c_1 \cos \theta + l \omega_x^2 \sin \theta) \cos \theta \\ \quad - (l c_1 \sin \theta + l \omega_x^2 \cos \theta + g) \sin \theta \\ \quad = h_2(\omega_x, \theta, a_y) = h_2(\mathbf{x}) \\ a_{bz} = -(a_y - l c_1 \cos \theta + l \omega_x^2 \sin \theta) \sin \theta \\ \quad - (l c_1 \sin \theta + l \omega_x^2 \cos \theta + g) \cos \theta \\ \quad = h_3(\omega_x, \theta, a_y) = h_3(\mathbf{x}) \end{cases} \quad (12)$$

The observation equation is shown as follows:

$$\begin{aligned}
\mathbf{z}_k &= \begin{bmatrix} \tilde{\omega}_x(k) \\ \tilde{a}_{by}(k) \\ \tilde{a}_{bz}(k) \end{bmatrix} = \begin{bmatrix} \omega_x(k) + n_{z\omega}(k) \\ h_2(\mathbf{x}_k) + n_{zaby}(k) \\ h_3(\mathbf{x}_k) + n_{zabz}(k) \end{bmatrix} \\
&= \begin{bmatrix} \omega_x(k) \\ h_2(\mathbf{x}_k) \\ h_3(\mathbf{x}_k) \end{bmatrix} + \begin{bmatrix} n_{z\omega}(k) \\ n_{zaby}(k) \\ n_{zabz}(k) \end{bmatrix} \\
&= h(\mathbf{x}_k) + \mathbf{n}_z(k)
\end{aligned} \tag{13}$$

where  $\mathbf{n}_z(k) = [n_{z\omega}(k) \ n_{zaby}(k) \ n_{zabz}(k)]^T$ .  $n_{z\omega}(k)$  is the measurement noise of the angular velocity  $\omega_x$ , and is considered to be a white noise.  $n_{zaby}(k)$  is the measurement noise of the acceleration  $a_{by}$ , and is considered to be a white noise.  $n_{zabz}(k)$  is the measurement noise of the acceleration  $a_{bz}$ , and is considered to be a white noise.

Taking into account the small amount of  $\theta$  and  $\omega_x$ , then  $h(\mathbf{x}_k)$  can be approximated to

$$h(\mathbf{x}_k) \approx \begin{bmatrix} \omega_x \\ a_y - lc_1 - g\theta \\ -l\omega_x^2 - a_y\theta - g \end{bmatrix}_{\mathbf{x}=\mathbf{x}_k} \tag{14}$$

where

$$c_1 \approx \frac{m_c l g}{m_c l + J_c} \theta + \frac{(m_c + m_w) l}{m_c l + J_c} a_y = c_{11} \theta + c_{12} a_y.$$

Then

$$\begin{aligned}
h(\mathbf{x}_k) &\approx \begin{bmatrix} \omega_x \\ a_y - l \frac{m_c g l \theta + (m_c + m_w) l a_y}{m_c l + J_c} - g\theta \\ -l\omega_x^2 - a_y\theta - g \end{bmatrix}_{\mathbf{x}=\mathbf{x}_k} \\
&= \begin{bmatrix} \omega_x \\ -(c_{11} l + g) \theta + (1 - c_{12} l) a_y \\ -l\omega_x^2 - a_y\theta - g \end{bmatrix}_{\mathbf{x}=\mathbf{x}_k}
\end{aligned} \tag{15}$$

So

$$\begin{cases} \mathbf{x}_k = f(\mathbf{x}_{k-1}) + \mathbf{n}_x(k) \\ \mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{n}_z(k) \end{cases} \tag{16}$$

### C. The Jacobi matrix for State Determination

Because the state equation and observation equation are all nonlinear, so the Extended Kalman Filter is used [5][6][7].

$$\mathbf{F}_k = \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}^T} \Big|_{\mathbf{x}=\hat{\mathbf{x}}_k} \text{ is the Jacobi matrix, so}$$

$$\mathbf{F}_k \approx \begin{bmatrix} 1 & Tc_{11} & Tc_{12} & 0 & 0 \\ T & 1+0.5T^2c_{11} & 0.5T^2c_{12} & 0 & 0 \\ -k_p T & 0 & 1 & 0 & 0 \\ 0 & 0 & T & 1 & 0 \\ 0 & 0 & 0.5T^2 & T & 1 \end{bmatrix} \tag{17}$$

and remains approximately a constant matrix.

$$\mathbf{H}_k = \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}^T} \Big|_{\mathbf{x}=\hat{\mathbf{x}}_k} \text{ is the Jacobi matrix, so}$$

$$\mathbf{H}_k \approx \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & -(c_{11} l + g) & 1 - c_{12} l & 0 & 0 \\ -2l\omega_x & -a_y & -\theta & 0 & 0 \end{bmatrix} \tag{18}$$

and remains a variable matrix.

## IV. THE TEST RESULTS

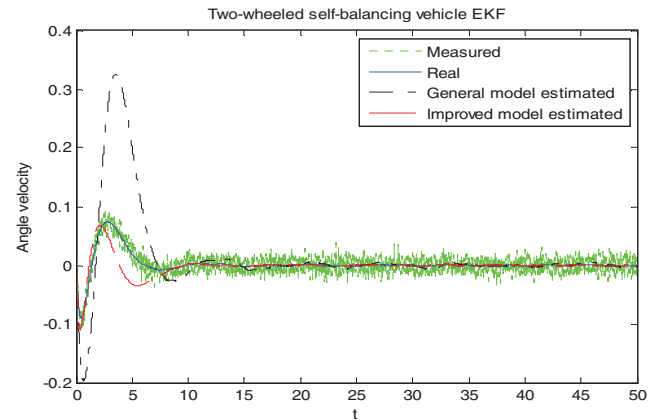
Based on the physical object we have established a parameter mathematical model of a two-wheeled self-balancing vehicle, and then do a mathematical simulation in the MATLAB platform.

Assume the two-wheeled self-balancing vehicle initially at rest, the initial values of the state variables are shown as follows:

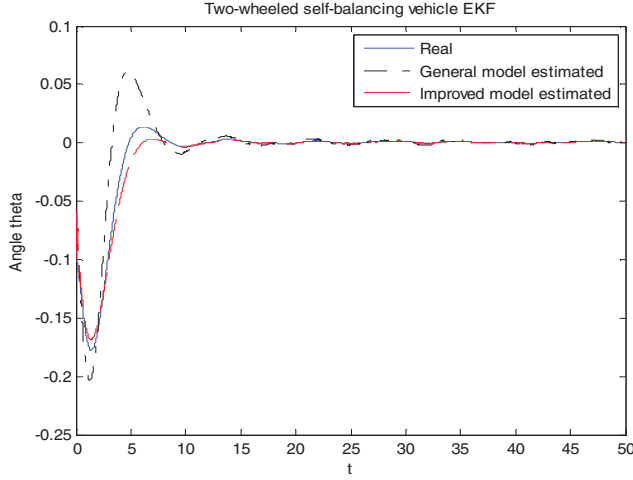
$$\begin{cases} \omega_x(0) = 0, \theta(0) = \arctg\left(\frac{a_{by}(0)}{a_{bz}(0)}\right) \\ a_y(0) = 0, v_y(0) = 0, p_y(0) = 0 \end{cases}$$

So the initial value of the attitude is set  $\theta = -0.1$ , other initial value of the state variables are set zero.

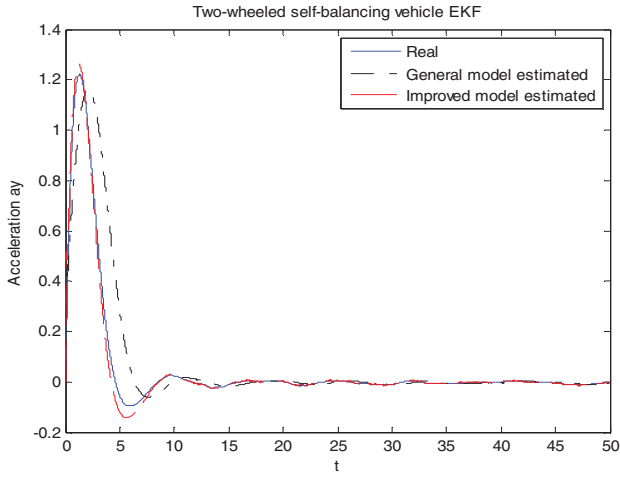
Test results of the improved computing model and the general model (regards the change of force as a noise) are shown as Fig. 3.



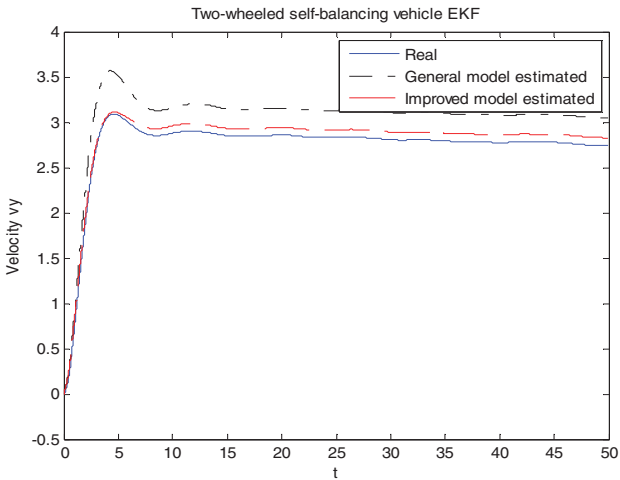
(a) The angular velocity  $\omega_x$



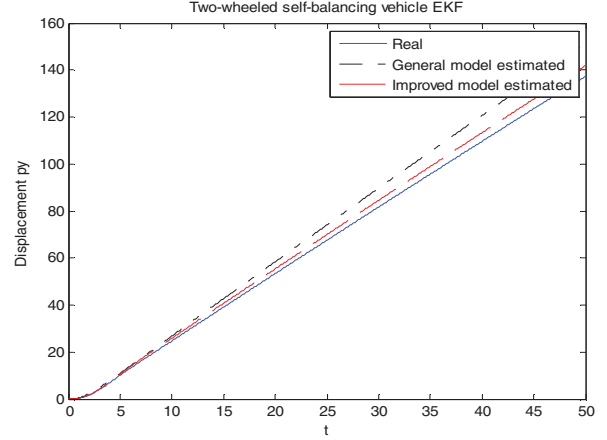
(b) The attitude angle  $\theta$



(c) The acceleration  $a_y$



(d) The velocity  $v_y$



(e) The displacement  $p_y$   
Fig. 3 Comparing of the state variables filtered

Fig. 3 (a) is the test result of the angular velocity  $\omega_x$ . Fig. 3 (b) is the test result of the attitude angle  $\theta$ . Fig. 3 (c) is the test result of the acceleration  $a_y$  in y-axis. Fig. 3 (d) is the test result of the velocity  $v_y$  in y-axis. Fig. 3 (e) is the test result of the displacement  $p_y$  in y-axis.

It can be seen from Fig. 3 (a), (b), (c), the estimating errors of the angular velocity  $\omega_x$ , the attitude angle  $\theta$  and acceleration  $a_y$  are relatively large in the transient status, but the estimating errors of the improved computing model are significantly better than the general model's estimating errors, whereas both estimating errors are relatively small in the steady status. From Fig. 3 (d), (e) the estimating velocity  $v_y$  and displacement  $p_y$  can be seen steady-status error are present, and the steady-status error of the estimating velocity  $v_y$  tends to a steady value, and the steady-status error of the estimating displacement  $p_y$  which is an integral of the velocity  $v_y$ , is increasing.

However, steady-status errors of the improved computing model are significantly less than the general model's steady-status errors. The estimating steady-status errors of the angular velocity  $\omega_x$ , the attitude angle  $\theta$  and the acceleration  $a_y$  are relatively small, because there are sensors correction (accelerometer, gyroscope), and the estimating steady-status errors of velocity  $v_y$  and displacement  $p_y$  are relatively large, because there is no sensor correction.

In summary, when employing the improved computing model, the estimating errors of the angular velocity  $\omega_x$ , the attitude angle  $\theta$  and the acceleration  $a_y$  are significantly reduced in the transient status, which can significantly improve the transient control effects because these three parameters are important state quantity of a two-wheeled self-balancing vehicle.

## V. CONCLUSIONS

When we change significantly the velocity of a

two-wheeled self-balancing vehicle, the vehicle does not run smoothly and fluctuates significantly for a long time. To address this problem, this paper presents an improved computing model based on the state variable of a two-wheeled self-balancing vehicle. Experimental results show that when employing the improved computing model, the estimating errors of the angular velocity  $\omega$ , the attitude angle  $\theta$  and the acceleration  $a$ , are significantly reduced in the transient status, which can significantly improve the transient control effects.

#### REFERENCES

- [1] F. Grasser, A. D'Arrigo, S. Colombi, and A. Rufer, "Joe: A mobile, inverted pendulum", *IEEE Trans. Industrial Electronics*, vol. 49, no. 1, pp. 107-114, 2002.
- [2] Tu Yunwu, Xu Junyan, Zhang Peiren, and Zhang Zhijian, "Model and Simulation of Self-balance Control System", *Journal of System Simulation*. Beijing, vol. 16, no. 4, pp.839-841, April 2004.
- [3] Tu Yunwu, "Study of Self-balance Control System", Dr. paper, Hefei: University of Science and Technology of China, 2004.
- [4] Katsuhiko Ogata, *Modern Control Engineering*, 3rd ed., Upper Saddle River: Prentice Hall, 2010, pp. 78-79.
- [5] Qin Yongyuan, Zhang Hongyue, and Wang Shuhua, *Principle of Kalman filter with integrated navigation*, Xi'an: Northwestern Polytechnical University Press, 1998, pp. 33-55.
- [6] J. L. Crassidis, F. L. Markley, and Y. Cheng. "A survey of nonlinear attitude estimation methods", *Journal of Guidance, Control, and Dynamics*, vol. 30, no. 1, pp. 12-28, 2007.
- [7] K. H. Kim, J. G. Lee, and C. G. Park. "Adaptive two-stage extended Kalman filter for a fault-tolerant INS-GPS loosely coupled system", *IEEE Transactions on Aerospace and Electronic Systems*, vol. 45, no. 1, pp. 125-137, January 2009.